

Each question is worth the number of marks indicated on the right and the total number of marks available is 65.

1. Let p and q be statement variables. Prove that

$$(p \vee q) \rightarrow (q \wedge \sim p) \equiv \sim p.$$

You may use either truth tables or logical equivalences. If you use logical equivalences, you should name the laws you use at each step

4
(4 marks)

Solution 1

$$\begin{aligned}
 & (p \vee q) \rightarrow (q \wedge \sim p) \\
 \equiv & \sim(p \vee q) \vee (q \wedge \sim p) && \text{by def. of } \rightarrow \\
 \equiv & (\sim p \wedge \sim q) \vee (q \wedge \sim p) && \text{by de Morgan's law} \\
 \equiv & (\sim p \wedge \sim q) \vee (\sim p \wedge q) && \text{by commutative law} \\
 \equiv & \sim p \wedge (\sim q \vee q) && \text{by Distributive law} \\
 \equiv & \sim p \wedge t && \text{by Negation law} \\
 \equiv & \sim p && \text{by Identity law}
 \end{aligned}$$

Solution 2

p	q	$(p \vee q) \rightarrow (q \wedge \sim p)$			$\sim p$
T	T	T	F	F	F
T	F	T	F	F	F
F	T	T	T	T	T
F	F	F	T	F	T

so
Same

$$(p \vee q) \rightarrow (q \wedge \sim p) \equiv \sim p$$

2. For each of the following statements, determine whether the statement is true or false. Provide a proof if the statement is true and provide a counterexample if the statement is false.

(6 marks)

- (a) For all $b, c, d \in \mathbb{Z}^+$, if $d \nmid (b - 2c)$ then $d \nmid b$ or $d \nmid c$.

True.

Proof Consider the contrapositive statement
 $\forall b, c, d \in \mathbb{Z}^+$, if $d \mid b$ and $d \mid c$ then $d \mid (b - 2c)$.

Suppose $b, c, d \in \mathbb{Z}^+$ and $d \mid b$ and $d \mid c$.

Then $b = ds$ for some $s \in \mathbb{Z}$ and $c = dt$ for some $t \in \mathbb{Z}$.

Hence $b - 2c = ds - 2(dt) = d(s - 2t)$.

Since $s, t \in \mathbb{Z}$, $s - 2t \in \mathbb{Z}$ so $d \mid (b - 2c)$.

$\therefore$ By contraposition

$\forall b, c, d \in \mathbb{Z}$, if $d \nmid (b - 2c)$ then $d \nmid b$ or $d \nmid c$.

- (b) For all $x, y \in \mathbb{Z}^+$, if $x \equiv y \pmod{5}$ then $2^x \equiv 2^y \pmod{5}$.

False.

Consider $x = 1$ and $y = 6$.

Then $x, y \in \mathbb{Z}^+$ and $1 \equiv 6 \pmod{5}$

but $2^1 \equiv 2 \pmod{5}$ and $2^6 \equiv 64 \equiv 4 \pmod{5}$

so $2^x \not\equiv 2^y \pmod{5}$.

3. Consider the sequence $\{a_n\}_{n \geq 1}$ defined recursively as

$$a_1 = 1, a_2 = 3, a_3 = 4 \text{ and } a_k = a_{k-1} + 2a_{k-3} \text{ for each integer } k \geq 4.$$

Use strong mathematical induction to prove that $a_n < 3^n$ for each integer $n \geq 1$.

(6 marks)

Proof Let $P(n)$ be the predicate $a_n < 3^n$.

Basis: Consider $n=1, 2$ and 3 .

$$a_1 = 1 < 3^1 = 3, \text{ so } P(1) \text{ is true.}$$

$$a_2 = 3 < 3^2 = 9, \text{ so } P(2) \text{ is true.}$$

$$a_3 = 4 < 3^3 = 27, \text{ so } P(3) \text{ is true.}$$

Ind. Hyp. Suppose $k \geq 3$ is an integer and $P(1), P(2), \dots, P(k)$ are true.

$$\text{Since } P(k) \text{ is true, } a_k < 3^k$$

$$\text{Since } P(k-2) \text{ is true, } a_{k-2} < 3^{k-2}$$

Consider $n = k+1$.

$$a_{k+1} = a_k + 2a_{k-2} \quad \text{by the rec. rel. since } k+1 \geq 4$$

$$< 3^k + 2 \cdot 3^{k-2} \quad \text{by the Ind. Hyp.}$$

$$= 3^{k-2}(3^2 + 2)$$

$$= 3^{k-2}(11)$$

$$< 3^{k-2}(27) \quad \text{since } 11 < 27$$

$$= 3^{k-2} \cdot 3^3$$

$$= 3^{k+1}$$

$$\therefore P(k+1) \text{ is true}$$

$\therefore$ By PSMI, $a_n < 3^n$ for each integer $n \geq 1$. $\square$

4

5. (a) Prove that for all sets A, B , and C that are subsets of a universal set,

$$A - (B \cup C) = (A - B) - C.$$

(3 marks)

$$\begin{aligned}
 A - (B \cup C) &= A \cap (B \cup C)^c && \text{by the set difference law} \\
 &= A \cap (B^c \cap C^c) && \text{by de Morgan's law} \\
 &= (A \cap B^c) \cap C^c && \text{by the associative law} \\
 &= (A - B) - C && \text{by the set difference law (twice)}
 \end{aligned}$$

- (b) Consider the sets $S = \{a, b\}$ and $T = \{a, c, d\}$. Indicate whether each of the following statements is true or false; no reason is needed.

(2 marks)

- (i) $S \subseteq T$ False
- (ii) $\emptyset \in \mathcal{P}(S)$ True
- (iii) $\emptyset \in T$ False
- (iv) $(a, a, a) \in S \times T$ False

5
0. Consider the function $f : \mathbb{Z}^+ \times \mathbb{Z}^+ \rightarrow \mathbb{Z}^+$ where $\forall (a, b) \in \mathbb{Z}^+ \times \mathbb{Z}^+, f((a, b)) = \text{lcm}(a, b)$.

(a) Is f one-one? Explain your answer.

(2 marks)

No.

$$(2, 4), (1, 4) \in \mathbb{Z}^+ \times \mathbb{Z}^+$$

$$f((2, 4)) = f((1, 4)) = 4$$

$$\text{but } (2, 4) \neq (1, 4)$$

so f is not one-one

(b) Is f onto? Explain your answer.

(2 marks)

Yes.

Consider $n \in \mathbb{Z}^+$

Then $(n, n) \in \mathbb{Z}^+ \times \mathbb{Z}^+$ and $\text{lcm}((n, n)) = n$

so f is onto.

6

7. Recall that $[-1, 3] = \{x \in \mathbb{R} : -1 \leq x \leq 3\}$ and that $(2, 4] = \{x \in \mathbb{R} : 2 < x \leq 4\}$. Define a function that demonstrates that $|[-1, 3]| \leq |(2, 4]|$. You must explain why it is a function and why it shows that $|[-1, 3]| \leq |(2, 4]|$.

6
(5 marks)

Consider $f: [-1, 3] \rightarrow (2, 4]$

where $f(x) = \frac{x}{3} + 3$

Since $-1 \leq x \leq 3$, $2\frac{2}{3} \leq \frac{x}{3} + 3 \leq 4$ and $(2\frac{2}{3}, 4] \subseteq (2, 4]$
and for each $x \in [-1, 3]$, the value $\frac{x}{3} + 3$ is unique.
Hence f is a function.

Also f is one-one.

Suppose $x_1, x_2 \in [-1, 3]$ and $f(x_1) = f(x_2)$.

Then $\frac{x_1}{3} + 3 = \frac{x_2}{3} + 3$ so $x_1 = x_2$.

$\therefore f$ is one-one.

Since f is a one-one function with domain $[-1, 3]$ and codomain $(2, 4]$ we have

$$|[-1, 3]| \leq |(2, 4]|.$$

7. Let S be the set of binary strings of length six. The weight of a binary string is the number of ones it contains, for example $\text{weight}(110110) = 4$. Define a relation ρ on S by

$$\forall s, t \in S, s \rho t \text{ if and only if } \text{weight}(s) \leq \text{weight}(t).$$

- (a) Prove that ρ is ^{neither} a partial order relation, ^{nor an equivalence relation}. (6 marks)
- (b) ρ is not antisymmetric.

For example $s = 010000$ and $t = 100000$.

Then $\text{weight}(s) = \text{weight}(t) = 1$ so $s \rho t$ and $t \rho s$ but $s \neq t$.

Hence ρ is not a partial order relation.

- (a) ρ is not symmetric.

For example $s = 100000$ and $t = 110000$.

Then $1 = \text{weight}(s) \leq \text{weight}(t) = 2$ so $s \rho t$ but $\text{weight}(t) \not\leq \text{weight}(s)$ so $t \not\rho s$.

Hence ρ is not an equivalence relation.

- (b) Prove that ρ is not a total order relation.

(1 mark)

9. Consider the group $(\mathbb{Z}_4 \times \mathbb{Z}_2, +)$.

(a) Prove that $\{(0,0), (0,1), (2,0), (2,1)\}$ is a subgroup of $(\mathbb{Z}_4 \times \mathbb{Z}_2, +)$.

4
(8 marks)

+	(0,0)	(0,1)	(2,0)	(2,1)
(0,0)	(0,0)	(0,1)	(2,0)	(2,1)
(0,1)	(0,1)	(0,0)	(2,1)	(2,0)
(2,0)	(2,0)	(2,1)	(0,0)	(0,1)
(2,1)	(2,1)	(2,0)	(0,1)	(0,0)

By the Cayley table we see the set is closed under addition.

$(0,0)$ is the identity of $\mathbb{Z}_4 \times \mathbb{Z}_2$ and also for this set under addition.

Each element has an inverse in the set.

$$(0,0)^{-1} = (0,0) \quad (0,1)^{-1} = (0,1) \quad (2,0)^{-1} = (2,0) \quad (2,1)^{-1} = (2,1)$$

$\therefore$ The set is a subgroup of $\mathbb{Z}_4 \times \mathbb{Z}_2$

$$(x,y) \quad \langle (x,y) \rangle$$

(b) Determine all the elements $x \in \mathbb{Z}_4 \times \mathbb{Z}_2$ for which $\langle x \rangle$ is isomorphic to $(\mathbb{Z}_4, +)$ and explain your reasoning.

Since the four elements considered in part (a) are all self-inverse, none of them will give a cyclic subgroup isomorphic to $\mathbb{Z}_4$. All the other elements of $\mathbb{Z}_4 \times \mathbb{Z}_2$ do give cyclic subgroups isomorphic to $(\mathbb{Z}_4, +)$. (3 marks)

(a) are all self-inverse, none of them will give a cyclic subgroup isomorphic to $\mathbb{Z}_4$. All the other elements of $\mathbb{Z}_4 \times \mathbb{Z}_2$ do give cyclic subgroups isomorphic to $(\mathbb{Z}_4, +)$.

$$\langle (1,0) \rangle = \{(1,0), (2,0), (3,0), (0,0)\} = \langle (3,0) \rangle$$

$$\langle (1,1) \rangle = \{(1,1), (2,0), (3,1), (0,0)\} = \langle (3,1) \rangle$$

9

10. (a) Determine the multiplicative inverse of 7 in the field $\mathbb{Z}_{11}$.

(2 marks)

Integers $\equiv 1 \pmod{11}$

12, 23, 34, 45, 56, ...

Since $56 = 7 \cdot 8$, 8 is the multiplicative inverse of 7.

Prove that

- (b) ~~Explain~~ why $\mathbb{Z}_{12}$ with addition and multiplication modulo 12 is not a field.

(2 marks)

$(\mathbb{Z}_{12} - \{[0]\}, \cdot)$ is not an abelian group.

$[3] \cdot [4] = [0]$ so the set is not closed under multiplication.

10. For each part (a), (b) and (c) of Question 10, you do not need to compute an integer answer, you may give a mathematical expression that answers the question.

In Queensland, driver licence identification numbers consist of a string of nine digits chosen from $\{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$ with repetition allowed. For the purposes of this question, assume that no other conditions are placed on the number, so any string of nine digits is possible.

- (a) If a driver licence identification number is chosen at random, what is the probability that it will have no repeated digits? (2 marks)

$$\begin{aligned} \text{No repeats} &: 10 \cdot 9 \cdot 8 \cdot 7 \cdot 6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 = 10! \\ \text{Total} &: 10^9 \\ \text{Probability} &= \frac{10!}{10^9} \end{aligned}$$

- (b) How many driver licence identification numbers contain at least two occurrences of the digit 1? (2 marks)

$$\begin{aligned} \text{Total} - (\# \text{ with no } 1) - (\# \text{ with one } 1) \\ = 10^9 - 9^9 - \binom{9}{1} \cdot 9^8 \\ = 10^9 - 2 \cdot 9^9 \end{aligned}$$

- (c) How many driver licence identification numbers contain exactly one even digit or exactly one digit that is divisible by 3? (Note that "or" is inclusive or.) (4 marks)

$$|A \cup B| = |A| + |B| - |A \cap B|$$

Let A be the set of ID#s containing exactly one even digit.

Let B be the set of ID#s containing exactly one digit divisible by 3.

Even 0, 2, 4, 6, 8 $\div$ by 3 0, 3, 6, 9 both 0, 6. neither 1, 5, 7

$$|A| = 5 \cdot \binom{9}{1} \cdot 5^8$$

$$|B| = 4 \cdot \binom{9}{1} \cdot 6^8$$

$$|A \cap B| = \begin{cases} \text{if contains 0 or 6} & 2 \cdot \binom{9}{1} 3^8 \\ \text{if not 0 or 6} & 3 \cdot \binom{9}{1} \cdot 2 \cdot \binom{8}{1} \cdot 3^7 \end{cases}$$

$$\therefore |A \cup B| = 9 \cdot 5^9 + 4 \cdot 9 \cdot 6^8 - (2 \cdot 9 \cdot 3^8 + 3 \cdot 9 \cdot 2 \cdot 8 \cdot 3^7)$$

- 11.** Determine the coefficient of x^5 in the expansion of $\left(3x^2 - \frac{2}{x}\right)^7$, giving your answer as an integer.

(3 marks)

$$\begin{aligned}
 \left(3x^2 - \frac{2}{x}\right)^7 &= \sum_{r=0}^7 \binom{7}{r} (3x^2)^{7-r} \left(-\frac{2}{x}\right)^r \\
 &= \sum_{r=0}^7 \binom{7}{r} 3^{7-r} (-2)^r x^{14-2r} x^{-r} \\
 &= \sum_{r=0}^7 \binom{7}{r} 3^{7-r} (-2)^r x^{14-3r}
 \end{aligned}$$

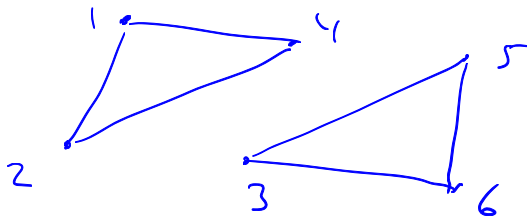
$$\begin{aligned}
 14 - 3r &= 5 \\
 r &= 3
 \end{aligned}$$

$$\begin{aligned}
 \text{Coeff } \binom{7}{3} 3^4 (-2)^3 \\
 &= \frac{7 \cdot 6 \cdot 5}{6} \cdot 81 \cdot (-8) \\
 &= 35 \cdot 81 \cdot (-8) \\
 &= -22680
 \end{aligned}$$

- 2
13. (a) Let G be the graph with the following adjacency matrix. Is G Eulerian? Explain your answer.

(2 marks)

$$\begin{bmatrix} 0 & 1 & 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 1 \\ 1 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 & 1 & 0 \end{bmatrix}$$



Not Eulerian
as not connected.

- (b) Let G be a simple, connected, Eulerian graph with eight vertices. Prove that the number of edges of G is greater than or equal to 8 but less than or equal to 24.

(4 marks)

Since G is connected and Eulerian
 $\deg(v) \geq 2$ for each vertex.

$$\therefore \sum \deg \geq 8 \cdot 2 = 16$$

So by the Handshake Theorem $\# \text{edges} \geq 8$.

Since G is simple $\deg(v) \leq 7$ for each vertex but since G is Eulerian
each vertex must have even degree
so $\deg(v) \leq 6$,

$$\therefore \sum \deg \leq 8 \cdot 6 = 48$$

so by the Handshake Theorem $\# \text{edges} \leq 24$.

END OF EXAMINATION

MATH1061/MATH7861 Examination Formula Page

Logical Equivalences

Given any statement variables p , q and r , a tautology $\mathbf{t}$ and a contradiction $\mathbf{c}$, the following logical equivalences hold.

<i>Commutative laws</i>	$p \wedge q \equiv q \wedge p$	$p \vee q \equiv q \vee p$
<i>Associative laws</i>	$(p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$	$(p \vee q) \vee r \equiv p \vee (q \vee r)$
<i>Distributive laws</i>	$p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$	$p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$
<i>Identity laws</i>	$p \wedge \mathbf{t} \equiv p$	$p \vee \mathbf{c} \equiv p$
<i>Negation laws</i>	$p \vee \sim p \equiv \mathbf{t}$	$p \wedge \sim p \equiv \mathbf{c}$
<i>Double negative laws</i>	$\sim(\sim p) \equiv p$	
<i>Idempotent laws</i>	$p \wedge p \equiv p$	$p \vee p \equiv p$
<i>Universal bound laws</i>	$p \vee \mathbf{t} \equiv \mathbf{t}$	$p \wedge \mathbf{c} \equiv \mathbf{c}$
<i>De Morgan's laws</i>	$\sim(p \wedge q) \equiv \sim p \vee \sim q$	$\sim(p \vee q) \equiv \sim p \wedge \sim q$
<i>Absorption laws</i>	$p \vee (p \wedge q) \equiv p$	$p \wedge (p \vee q) \equiv p$
<i>Negations of $\mathbf{t}$ and $\mathbf{c}$</i>	$\sim \mathbf{t} \equiv \mathbf{c}$	$\sim \mathbf{c} \equiv \mathbf{t}$

Set Identities For all sets A , B and C that are subsets of a universal set U :

1. Commutative Laws (a) $A \cup B = B \cup A$ (b) $A \cap B = B \cap A$
2. Associative Laws (a) $(A \cup B) \cup C = A \cup (B \cup C)$
(b) $(A \cap B) \cap C = A \cap (B \cap C)$
3. Distributive Laws (a) $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$
(b) $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$
4. Identity Laws (a) $A \cup \emptyset = A$ (b) $A \cap U = A$
5. Complement Laws (a) $A \cup A^c = U$ (b) $A \cap A^c = \emptyset$
6. Double Complement Law $(A^c)^c = A$
7. Idempotent Laws (a) $A \cup A = A$ (b) $A \cap A = A$
8. Universal Bound Laws (a) $A \cup U = U$ (b) $A \cap \emptyset = \emptyset$
9. De Morgan's Laws (a) $(A \cup B)^c = A^c \cap B^c$ (b) $(A \cap B)^c = A^c \cup B^c$
10. Absorption Laws (a) $A \cup (A \cap B) = A$ (b) $A \cap (A \cup B) = A$
11. Complement of U and $\emptyset$ (a) $U^c = \emptyset$ (b) $\emptyset^c = U$
12. Set Difference Law $A - B = A \cap B^c$

The Quotient-Remainder Theorem Given any integer n and any positive integer d , there exist unique integers q and r such that $n = dq + r$ and $0 \leq r < d$.

Unique Factorisation Theorem Given any integer $n > 1$, there exists a positive integer k , distinct prime numbers $p_1, p_2, \dots, p_k$, and positive integers $e_1, e_2, \dots, e_k$ such that $n = p_1^{e_1} p_2^{e_2} \cdots p_k^{e_k}$, and any other expression for n as a product of prime numbers is identical to this except perhaps for the order in which the factors are written.

Schröder-Bernstein Theorem For all sets A and B , if $|A| \leq |B|$ and $|A| \geq |B|$ then $|A| = |B|$.

Binomial Theorem Given any real numbers a and b , and any nonnegative integer n ,

$$(a + b)^n = \sum_{k=0}^n \binom{n}{k} a^{n-k} b^k.$$